

ASTR 301: Modern Astrophysics and Stellar Structure

Term: Fall 2026 | Instructor: Dr. S. Chandrasekhar (chandra@astro.univ.edu)

Course Description & Overview:

Thermodynamics of stellar interiors, radiative transfer, nuclear nucleosynthesis, degenerate matter, and relativistic gravitational collapse.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
Stellar Spectroscopy Lab	25%	Week 04	Observational Data Analysis
Hydrostatic Equilibrium Numerical Model	30%	Week 08	Python Solver & Writeup
Final Observational Research Project	35%	Week 12	AAS Format Research Paper
Astrophysical Problem Sets	10%	Week 10	Calculus Problem Sets

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: Blackbody Radiation & Magnitudes	Carroll & Ostlie (Ch. 3)
Week 02	Module 02: Spectral Classification & HR Diagram	Carroll & Ostlie (Ch. 8)
Week 03	Module 03: Stellar Atmospheres & Radiative Transfer	Rybicki & Lightman (Ch. 1)
Week 04	Module 04: Equations of Stellar Structure	Hansen et al. (Ch. 1–3)
Week 05	Module 05: Nuclear Fusion: p-p Chain and CNO	Bethe (1939); Clayton (Ch. 4)
Week 06	Module 06: Opacity & Convective Energy Transport	Kippenhahn & Weigert (Ch. 5–7)
Week 07	Module 07: Main Sequence & Post-Main Evolution	Carroll & Ostlie (Ch. 13)
Week 08	Module 08: White Dwarfs & Degeneracy Limits	Chandrasekhar (1931)
Week 09	Module 09: Supernovae Core Collapse Nucleosynthesis	Woosley et al. (2002)
Week 10	Module 10: Neutron Stars & Pulsar Mechanics	Shapiro & Teukolsky (Ch. 10)
Week 11	Module 11: General Relativity & Black Holes	Misner, Thorne & Wheeler (Ch. 31)
Week 12	Module 12: Cosmic Microwave Background & Review	Planck Collaboration (2018)